

EFFiton : Design & Implementation of Sample Color Changes using Spectrophotometry & Implementation Design for Casing

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Abstract—Precise measurement and analysis of color changes in samples are crucial requirements for color degradation detection that is used for wastewater treatment. Spectrophotometry is an analytical method used to measure the absorbance and transmittance of light by a colored sample. This paper presents the design and implementation of one of the subsystems in EFFiton: a portable, compact, and low-cost spectrophotometer. This system features a 6V halogen lamp, a convex lens, a 600 lines/mm diffraction grating, two mirrors, and a TSL1401CL sensor. Data acquisition and processing from the TSL1401CL sensor are performed using an ESP32 microcontroller, with the resulting spectrum visualized on the EFFiton interface layer.

Given the need to develop a device that capable of performing degradation with multiple procedures in a single unit, this paper also discusses the implementation of a casing to integrate the entire EFFiton subsystem into a single unit. To that end, this paper also discusses the implementation of the casing design to produce a high-quality product.

Keywords—Spectrofotometry, TSL1401CL, Diffraction Grating, Halogen Lamp, Casing.

I. INTRODUCTION

While Indonesia's textile industry significantly boosts the national economy and provides millions of jobs [1], its manufacturing processes generate highly toxic, non-biodegradable wastewater laden with synthetic dyes and heavy metals. This persistent, strongly colored effluent severely damages aquatic ecosystems by blocking sunlight, which hinders photosynthesis and depletes dissolved oxygen [2], while also posing long-term carcinogenic risks to local communities. To mitigate these impacts and protect the ecosystem as illustrated in Figure 1, our final-year project focuses on secondary-stage wastewater treatment by utilizing adsorption to remove organic dyes and dissolved pollutants, specifically testing samples of Methylene Blue and red food coloring.

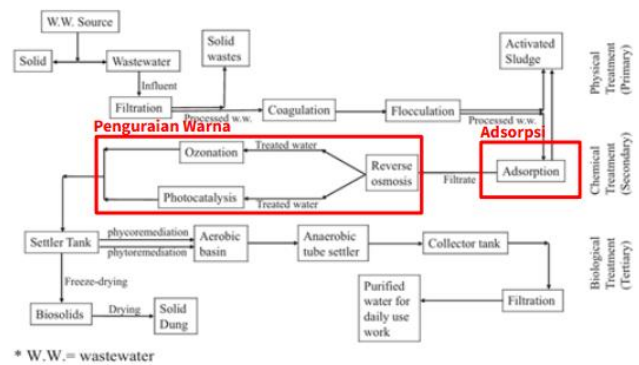


Figure 1 : Textile Waste Processing Process

Therefore, we developed a device called EFFiton to address this issue. With this device, we first perform chemical treatment through an adsorption process using zeolite. Then, we carry out the color removal process using one of the Advanced Oxidation Processes (AOPs), namely the Electro-Fenton process. So, part of our system involves treating wastewater in the secondary treatment stage, specifically in the adsorption and decolorization processes, using the Electro-Fenton method.

Therefore, rigorous and continuous monitoring in the electro-fenton process is an absolute necessity to ensure the presence of to determine whether this reaction causes degradation or color changes became a clear water. Detecting color changes and analyzing the spectral profile of liquid samples are fundamental steps in evaluating the efficiency of wastewater treatment processes. While these tasks are crucial across a wide range of applications—from environmental monitoring and chemical reaction analysis to biological instrumentation—conventional monitoring methodologies remain highly restrictive. Commercial instruments, such as standard laboratory benchtop spectrophotometers, provide high precision but fall short of meeting the practical demands of field monitoring. They are generally bulky, expensive, require controlled environments, and fundamentally lack the portability needed for rapid, *in-situ* analysis. The system created in EFFiton is as follows listed in Figure 2.

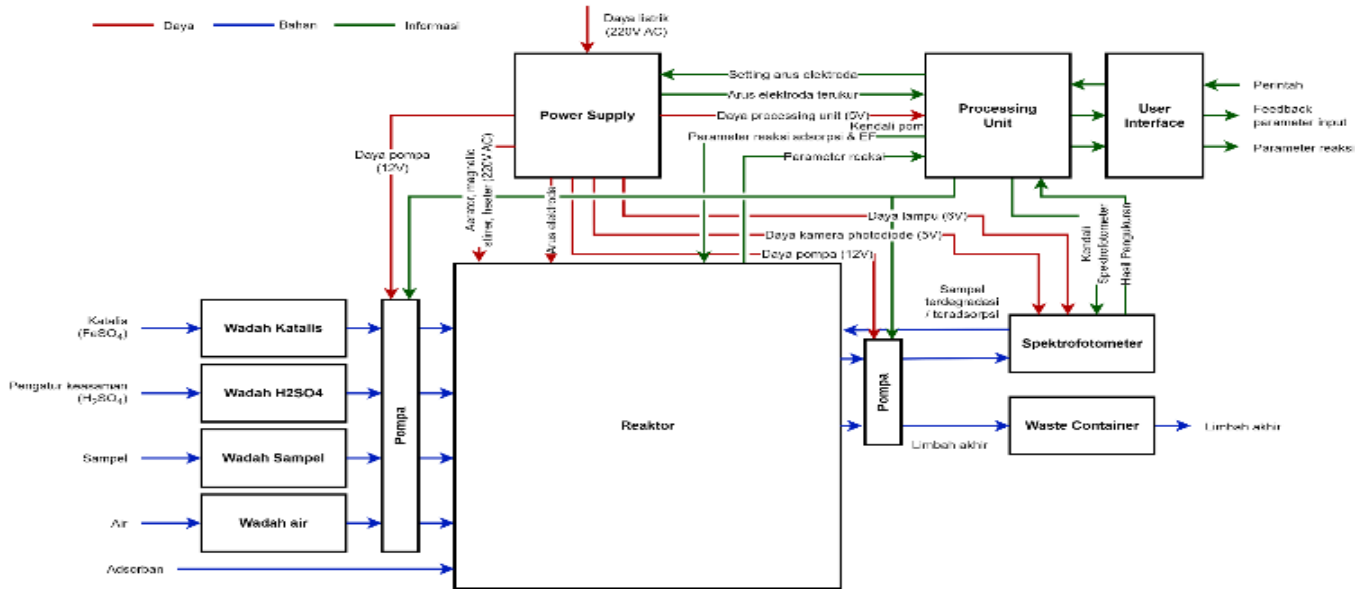


Figure 2 : Level 1 System of EFFiton

This paper discusses a subsystem of spectrophotometry, the design of the spectrophotometer used to measure changes in color degradation caused by the Electro-Fenton process in a reactor. Using this spectrophotometer, we can observe differences in color degradation in the sample before the process, after absorption, and after the Electro-Fenton process. Due to the limitations of laboratory spectrophotometers, which are not portable, color degradation changes cannot be monitored in a step-by-step process using a single instrument.

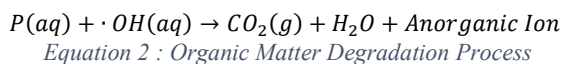
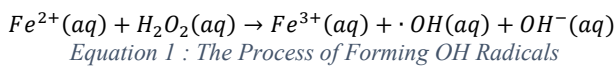
Therefore, EFFiton was designed to address these critical limitations by offering a highly portable spectrophotometer design that effectively utilizes microcontroller and optoelectronic components. This paper will discuss the hardware architecture, optical design, and sensor readings used to identify color changes through light spectrum profiles, as well as a comparison of the local instrument's specifications with those of commercial spectrophotometers.

II. LITERATURE REVIEW

The spectrophotometry system consists of a light source compartment, a sample cuvette holder, an optical dispersion system, and a digital signal processing unit to get the analog sensor value.

A. Electro-Fenton Process

Electro-Fenton process is an electrochemical advanced oxidation process based on in situ generation of hydroxyl radicals $\cdot OH$ by electrochemistry, unlike the chemical process Fenton process that proceeds with added reagents [4].



B. Optical System and Dispersion

The light source used as the polychromatic light source is a halogen lamp. We used a 6V 10W halogen lamp, which is commonly used for optical applications. We chose a halogen lamp because, as shown in Figure 1, the light output from a halogen lamp is sufficiently stable within the UV-Vis range we intend to use for measurements, namely 400–800 nm. This ensures that there will be no significant spikes that could affect the measurements.

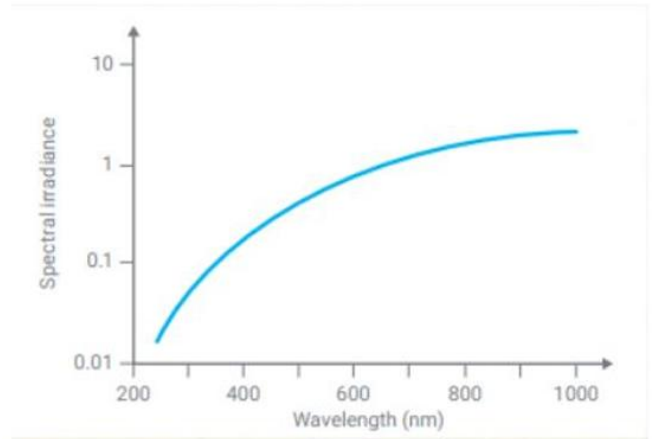


Figure 3 : Spectral Irradiance Halogen Lamp

C. TSL1401CL Sensor

To capture light, data must be captured according to the timing specified by the sensor. This allows the microcontroller to be synchronized with the device via the CLK pin, which then sends a signal to SI to start image capture, so that A0 can send an analog signal to the microcontroller for processing. Since each clock cycle corresponds to one pixel quantization, this process is performed over 128 clock cycles. Therefore, this needs to be adjusted in the code.



Figure 4 : TSL1401CL

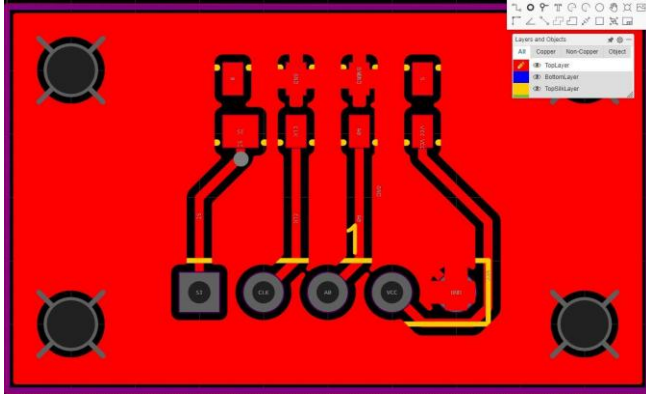


Figure 5 : Printed PCB for TSL1401CL

III. DESIGN & IMPLEMENTATION

A. Spectrophotometry Design & Implementation

The spectrophotometry system consists of a light source compartment, a sample cuvette holder, an optical dispersion system, and a digital signal processing unit to get the analog sensor value. This can be described as follows:

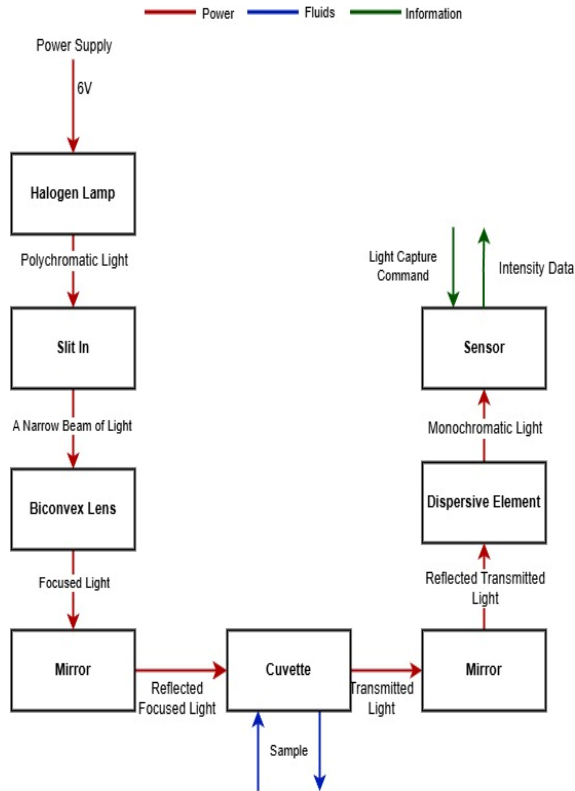


Figure 6 : Spectrophotometry's Block Diagram

The lamp serves as a source of polychromatic light, which is then directed through a slit onto a lens. The lens is then directed toward the cuvette using a mirror. The cuvette, which contains the sample, absorbs certain parts of the light spectrum from the lamp's polychromatic light. Next, the polychromatic light that has passed through the cuvette will be directed toward a dispersive element. This dispersive element will then break down the polychromatic light into monochromatic light with a specific wavelength range. After that, the monochromatic light will be directed precisely toward the sensor to determine the sample's absorbance and which specific wavelengths are being absorbed by the cuvette.

From the block diagram, spectrophotometry was implemented by taking into account the calculations required for the design. According to the specifications, the visible light spectrophotometer has a range of 400 nm – 800 nm. Therefore, $\lambda_1 = 400 \text{ nm}$, $\lambda_2 = 800 \text{ nm}$. We want the light from the first diffraction order—that is, the first-order diffraction—to obtain the brightest signal, ensuring more accurate sensor readings, and to achieve the smallest diffraction angle for easier positioning. So, we used $n = 1$ with dispersive element specifications 600 lines/mm and $d = \frac{1}{600 \text{ lines/mm}} \approx 1.667 \times 10^{-6} \text{ m}$.

Given these values, we need to determine the angle required by the dispersive element so that the resulting dispersed light is directed precisely toward the sensor. The required angle can be calculated as follows :

$$\sin \theta = \frac{n\lambda}{d}$$

Substituting those values with $n = 1$ and the values of d yields the angle at θ_1 and θ_2 is : 13.89 and 28.69.

Now, we need to determine the distance between the dispersive element and the sensor so that it hits the sensor accurately. Therefore, by knowing the sensor's length from the datasheet $L = 9.4 \text{ mm}$ and the distance between the pixels and the front of the sensor $s = 0.25 \text{ mm}$. Maka, dapat dihitung nilainya adalah sebagai berikut :

$$L = 2R \times \sin\left(\frac{\theta}{2}\right)$$

$$R = 48.01 \text{ mm}$$

Taking into account the 0.25 mm protrusion of the sensor, the distance between the dispersive element and the sensor is :

$$R - s = 48.01 - 0.25 = 47.76 \text{ mm}$$

For the halogen lamp assembly, a rear cover is provided to test the accuracy of the glass placement. Beneath it is a 5V DC fan that dissipates heat from the spectrometer to prevent the halogen lamp from overheating the system. In addition, a 5-cm gap was left between the lens and the slit because the lens is designed to focus light at a distance of approximately 5 cm.

IV. RESULT AND ANALYSIS

A. Spectrophotometry Results

The results of the implemented spectrophotometry design will be divided into three categories: light reflection accuracy, spectrophotometry calibration, and spectrophotometry integration. The section on light reflection accuracy will discuss how to achieve precise light placement through iterative adjustments to the specifications of the slit, the glass, and the position of the dispersive element.

1. Light Reflection Accuracy Testing

For this test, we verified that the light reflection was as intended and ensured that the sensor could detect a difference in the intensity of the light reaching it. First, to ensure that the halogen lamp would be properly aligned with the cuvette and directed toward the dispersive element, we conducted a test using a laser. The results are as follows:



Figure 10 : Implementation with Laser

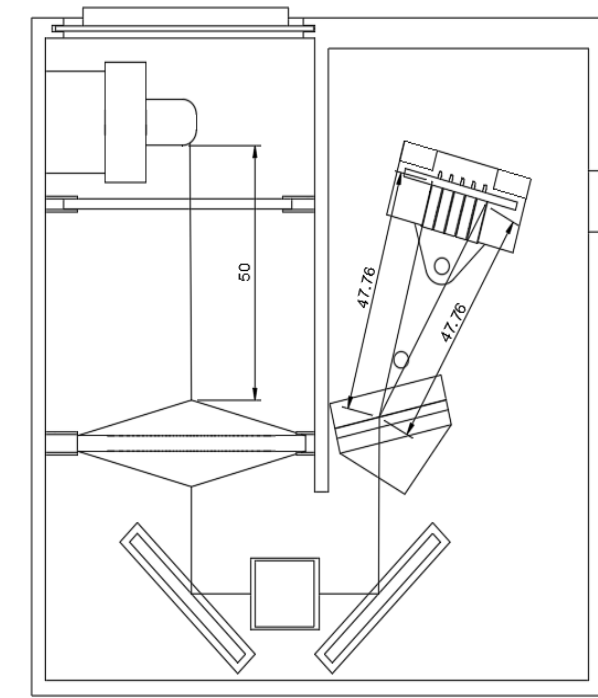


Figure 7 : Drawing Results for the Design



Figure 8 : Implemented Design

B. Casing Design

For the enclosure, the design we propose for this project is shown in Figure 7. For the frame, we are using 20x20 aluminum extrusions because they are easy to connect using hardware such as T-nuts, corner brackets, and other components. The exterior is covered with 2mm-thick white acrylic.

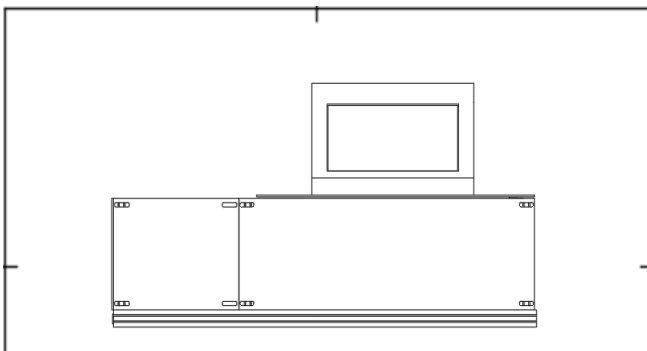


Figure 9 : Casing Design



Figure 11 : Implementation Using a Halogen Light Source

The test results show that the dispersive element successfully directs the monochromatic light spectrum precisely onto the sensor. However, during implementation, several factors affect this accuracy. A glass component in the

design shifts slightly when subjected to vibrations, causing the light beam to shift slightly upward or downward.

2. Spectrophotometry Calibration

Spectrophotometry is calibrated in two ways. The first is to test the accuracy of the spectrophotometric spectral results. The goal is to verify whether the displayed spectral values match the reference values obtained from measurements using a standard laboratory spectrophotometer. The second is to test the linearity of the spectrophotometer. Linearity is tested using a method involving 5 samples of Methylene Blue. Each sample is given 1 dropper of Methylene Blue and mixed with water at a 1:1 dilution ratio. From the data readings, it is expected that the spectrophotometer will produce a linear graph.

The spectrophotometric calibration method involved preparing two samples: one with Methylene Blue and the other with red food coloring. These two samples were then used to compare the spectrophotometric calibration for the EFFiton with the standard laboratory spectrophotometer during testing at the PPNN laboratory.

For the Methylene Blue sample, one pipette drop was diluted with 20 mL of water. Meanwhile, for the red food coloring sample, 2 drops of food coloring are mixed with 20 mL of water. However, during testing, the food coloring was deemed too concentrated for the laboratory spectrophotometer, resulting in a very high absorbance. Therefore, the food coloring was diluted further in a 1:1 ratio. The results of the two samples, measured using standard laboratory spectrophotometry, are as follows:

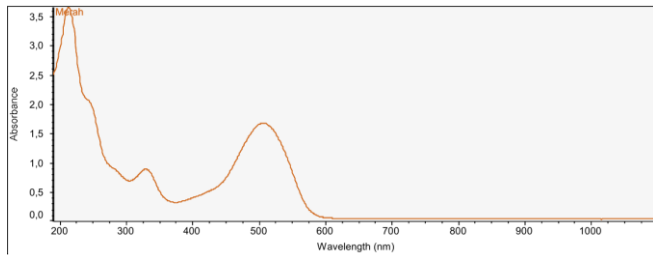


Figure 12 : Red Dye Measurement Results Using Laboratory Spectrophotometry

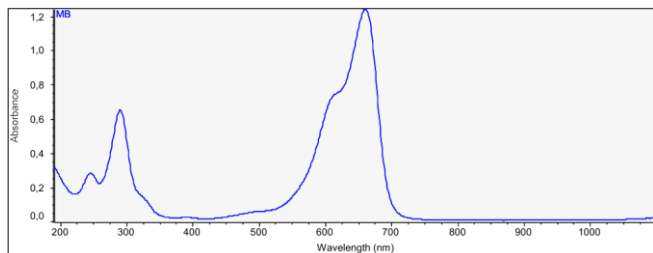


Figure 13 : Methylene Blue Measurement Results Using Laboratory Spectrophotometry

As shown in Figure 9 & Figure 10, Methylene Blue produces a peak at 661 nm. In contrast, the red food dye shows two peaks, at approximately 225 nm and 507 nm. Since the 225 nm peak falls within the UV spectrum, we will disregard it and focus only on the 507 nm peak.

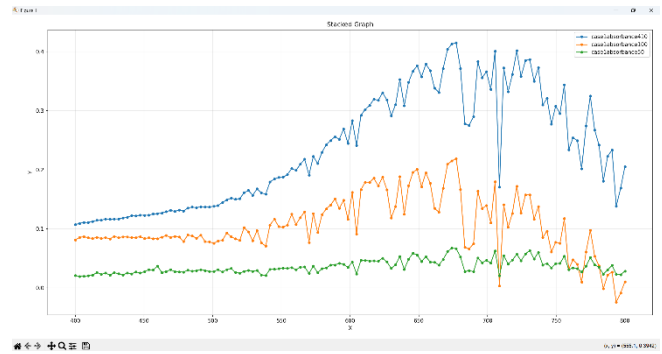


Figure 14 : Spectral Testing with Methylene Blue

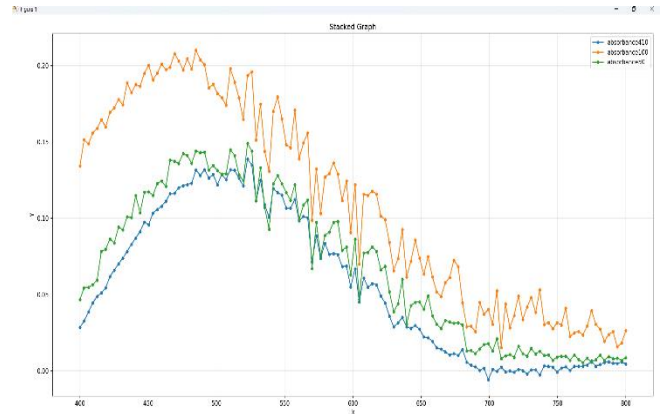


Figure 15 : Spectral Testing with Red Dye

In this testing phase, we made a minor error by reading the mapping in reverse. However, when the mapping was reversed and the wavelength scale was inverted, a peak was observed for Methylene Blue at approximately 650 nm and for the red dye at approximately 482 nm. Upon calculation, the difference between the laboratory spectrophotometer and the spectrophotometer we designed is:

$$\Delta_{error(MB)} = \left| \frac{650 - 661}{661} \right| \times 100\% = 1.67\%$$

$$\Delta_{error(red)} = \left| \frac{482 - 507}{507} \right| \times 100\% = 4.9\%$$

It should also be noted that the specifications of our spectrophotometer have a higher resolution compared laboratory spectrophotometers. According to the data provided by the laboratory, the resolution of the laboratory spectrophotometer is 1 nm. Meanwhile, the resolution of the spectrophotometer we built is higher (3).

$$resolution = \frac{\lambda_{max} - \lambda_{min}}{pixels} = \frac{800 - 400}{128} = 3.128 \text{ nm/pixel}$$

Equation 3 : Spectrophotometry Resolution (3)

In other words, 1 pixel value from the TSL1401CL sensor represents 3.128 nm. Since laboratory spectrophotometry of the resulting data yields a resolution of 1 nm/pixel, it is clear that this factor can also affect the Δ_{error} . Thus, the fact that the resolution is 3.12 times lower yet still remains below 5% indicates that the performance is quite good.

Sample's Diluent Volume	Absorbance
20 mL	0.0730
16 mL	0.1467
12 mL	0.2641
8 mL	0.3185
4 mL	0.3586

Table 1 : Absorbance Data for Linearity Test

For the linearity test, the graph of the five samples is shown in Figure 14. We selected the peak of Methylene Blue at specific points and plotted them on the graph to determine the linear regression and calculate the correlation coefficient. The method used to calculate the regression coefficient was the Pearson model (4).

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Equation 4 : Correlation Coefficient Pearson Model (4)

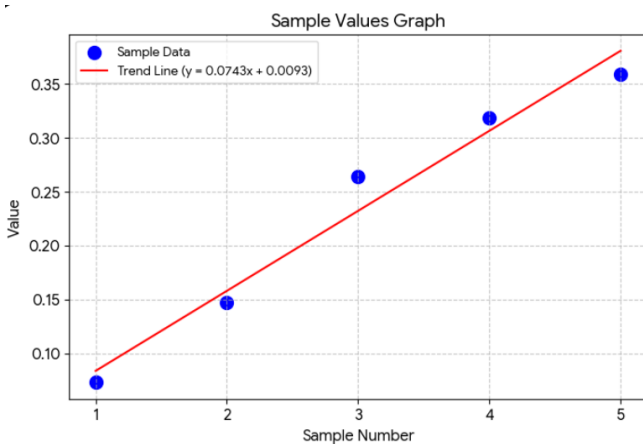


Figure 16 : Absorbance Graph for Linearity Test

Ref	Regression Coefficient	Error
[5]	0.98	± 5.7%
[6]	0.9921	± 3.34% at 100 ppm ± 5.43% at 200 ppm

Table 2 : Reference Table

The value of r was found to be 0.9833. The trend line has the equation $y = 0.0743x + 0.0093$. When we compare our results with those from other spectrophotometric references, we see that our regression coefficient and error are still within an acceptable range—specifically, they fall below the threshold—so the data obtained is considered sufficiently reliable.

3. Spectrophotometry Integration

In this section, spectrophotometry was tested using a method that integrates it with other subsystems and with specific halogen lamp timing to determine whether there are differences in readings when the system is combined versus when it is separated.



Figure 17 : Spectrophotometric Results with Subsystem Integration

When we performed the spectrophotometric integration, we found that part of the spectrophotometer had shifted because the printed PLA component was not stable enough to hold the mirror in place. This was because when we purchased the mirror online, the shape of the mirror we received did not match the specifications. When we attempted to cut the glass with a glass cutter, we were unable to achieve a clean cut according to our specifications because there was a section of the glass that was extremely dense and could not be cut. As a result of the integration, the data from the Red Dye spectrophotometry showed a peak at 660 nm, and the Methylene Blue data showed a peak around 725 nm.

V. CONCLUSION

Based on calibration using a standard laboratory spectrophotometer, the device has a relatively low error rate : 1.67% for Methylene Blue and 4.9% for Red Food Dye Samples. Based on the linearity test, the system produced a correlation coefficient (r) of 0.9833 from the absorbance values, indicating a strong correlation between the variables. The conceptual design and functionality is already proven good, but improvements are still needed regarding the stability of mechanical component.

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